

Prerequisite Diagnostic Quiz — Solutions

M 375T Fall 2026 — 2 pts per question.

1. $Av = \begin{pmatrix} 2 \cdot 1 + 1 \cdot (-1) \\ 0 \cdot 1 + 3 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}.$
2. Subtract $3 \times$ (first eq.) from the second: $-2y = -4 \Rightarrow y = 2$, then $x = 1$.
3. $Mv = \begin{pmatrix} 3+1 \\ 1+3 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4v$, so v is an eigenvector with eigenvalue $\lambda = 4$. (*Full credit requires the check $Mv = 4v$, not just the number 4.*)
4. $\mathbb{P}(\text{sum} = 12) = \frac{1}{36}$ (only 6+6). $\mathbb{P}(\text{sum} \geq 11) = \frac{3}{36} = \frac{1}{12}$ (5+6, 6+5, 6+6).
5. $\mathbb{E}[X] = -\frac{1}{2} + \frac{1}{4} + \frac{3}{4} = \frac{1}{2}$. $\mathbb{E}[X^2] = \frac{1}{2} + \frac{1}{4} + \frac{9}{4} = 3$, so $\text{Var}(X) = 3 - \left(\frac{1}{2}\right)^2 = \frac{11}{4}$.
6. (a) $\mathbb{P}(Z > 0) = \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{1}{2}$, by symmetry of φ about 0 and $\int_{-\infty}^\infty \varphi = 1$. (*Full credit requires the integral with the correct limits, not just the number.*)
(b) Mean = 1, variance = $2^2 \cdot 1 = 4$. (*Common slip: variance 2. One point total if the integral is right but the variance is wrong, or vice versa.*)